

Nicolas Alvarez Arce

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EDUCATION

Loyola Marymount University

Los Angeles, CA

Bachelor of Science in Mechanical Engineering, Minor in Math

May 2025

- Relevant Coursework: Computer-Aided Design (CAD), Algorithms & Applications (MATLAB & Python), Heat Transfer, Thermodynamics, Finite Element Analysis, Rapid Prototyping, Robotics, Aerodynamics

EXPERIENCE

Walltopia

Sofia, Bulgaria

Mechanical Engineering Intern

Jul 2025 – Present

- Performed load verification and EN 12572 / EN 13814 compliance for a new product line and reviewed CAD across 3 climbing-wall and Rollglider projects (including the Georgia Aquarium).
- Audited, cleaned, and merged after-sales datasets and re-engaged inactive clients.

Boeing — Materials Lab

Los Angeles, CA

Student Research Assistant

Nov 2022 – May 2025

- Ran hardness and tensile testing on 30+ additively manufactured AlSi10Mg samples, mapping a peak-aging strength curve to refine Boeing post-processing standards; co-authoring a manuscript under peer review.
- Characterized microstructure via optical microscopy and SEM; presented findings to 20+ engineers at ASME meetings.

LMU Fluids of Astrophysical Bodies (FAB) Lab

Los Angeles, CA

Mechanical Engineer

Aug – Dec 2023

- Validated (FEA, 3× safety factor) and machined a ~1 m rotary convection table for icy-moon planetary-science research, taking it from CAD through fabrication with manufacturability reviews.
- Developed the concept design for a cryogenic "expanding-box" apparatus enabling a research proposal submitted to NASA JPL.

LMU Thermal Fluids Lab

Los Angeles, CA

Student Research Assistant

May – Aug 2023

- Machined a thermocouple-embedded copper test surface (± 1 °C control) and ran 1,000+ droplet-impact trials across water and isopropanol over a 30–80 °C range to study evaporative cooling for electronics.
- Processed LabVIEW/DAQ data in MATLAB to characterize droplet residence time and heat-transfer behavior.

Ruhrpumpen

Monterrey, Mexico

Project Engineer Intern

May – Oct 2024

- Cut cycle times up to 15% through time studies and Lean Six Sigma, and diagnosed a recurring casting defect, standardizing a repair that improved efficiency ~10%.
- Managed purchase orders, inspection plans, and supplier documentation to API/ASME/ISO compliance and 100% on-time delivery across a 4-pump high-pressure program.

SKILLS

SolidWorks • Abaqus • GD&T (ASME Y14.5) • MATLAB • Python • C/C++ • Arduino • FEA • Machining • Prototyping • 3D Printing (FDM/SLA) • Lean Six Sigma Green Belt • Prompt Engineering & AI Workflows (Claude, Gemini)

Projects: Autonomous Quadcopter (1st-Place, Flight Controls); IGVC Autonomous Ground Vehicle (Capstone); Rocket Fin FEA (Abaqus); Two-Stage Rocket (Propulsion, Tripoli L1/L2); 6-DOF Stewart Platform; Kitesurfing Action-Cam Mount.

Leadership: President, LMARS (Rocketry); Treasurer, Climbing Club (300+ members); D1 Men's Rowing (Coxswain); Program Transition Lead, LMU Rowing.

Certifications: CSWA • Lean Six Sigma Green Belt • Google Data Analytics • Anthropic AI Fluency • Tripoli L1/L2

Languages: Spanish (native) • English (fluent)